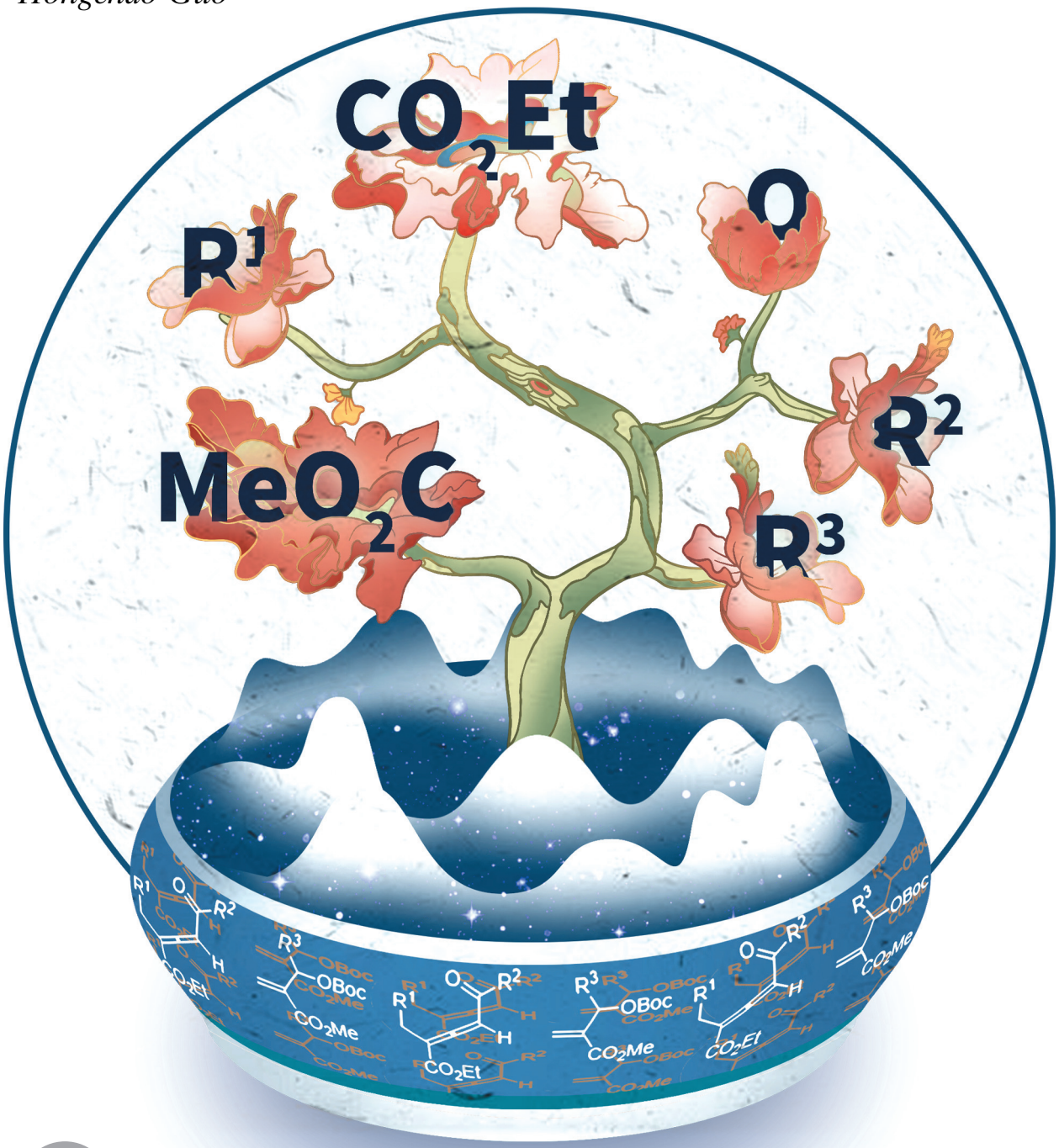
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VIP Organocatalysis | Very Important Paper

Organocatalytic Asymmetric C(sp^2)–H Allylic Alkylation: Enantioselective Synthesis of Tetrasubstituted Allenoates

*Yimin Hu, Wangyu Shi, Bing Zheng, Jianning Liao, Wei Wang, Yongjun Wu, and Hongchao Guo**



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Abstract: *Herein we describe the first organocatalytic asymmetric C(sp^2)–H allylation of racemic trisubstituted allenoates with Morita–Baylis–Hillman (MBH) carbonates to access axially chiral tetrasubstituted allenoates. Various trisubstituted allenoates and MBH carbonates were well tolerated under mild reaction conditions, providing novel chiral tetrasubstituted allenoates with adjacent axial chirality and tertiary carbon stereocenters in high yields with good to excellent diastereoselectivities and enantioselectivities.*

Asymmetric allylic alkylation is one of the most useful carbon-carbon bond construction tools in synthesis of natural products and biologically valuable molecules.[1] It displayed the powerful ability to transform achiral, prochiral, or chiral racemic material to enantiopure compounds. Asymmetric allylic alkylation reactions such as Tsuji–Trost allylation reactions via π-allyl species,[2] and organocatalytic allylic alkylation of modified Morita-Baylis–Hillman (MBH) adducts[3] have thus intensely been studied. In general, in Tsuji–Trost allylation reactions and organocatalytic allylic alkylations, active methylene compounds having one or two electron-withdrawing groups, enamines and enolates act as the soft carbon nucleophiles (Scheme 1 a and b). Asymmetric process occurs at highly activated sp^3 center of these soft carbon nucleophiles, on which the proton has certain acidity, leading to asymmetric allylation of activated C(sp^3)-H functional group (Scheme 1 a and b). To the best of our knowledge, asymmetric C(sp^2)-H allylation has not been reported in Tsuji–Trost allylation reactions and organocatalytic allylic alkylations. This triggered our keen interest, and inspired us to investigate application of the electron-deficeint allenes in these asymmetric allylation reactions, which may provide a new approach to synthesize chiral allenes.

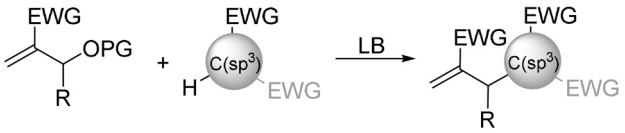
Axially chiral allenes are a key structural moiety in many natural products, bioactive molecules and chiral catalysts.[4] New catalytic asymmetric reactions for their synthesis, especially for synthesis of chiral molecules with multiple elements of chirality by simultaneously controlling two different elements of stereogenicity,[5] are highly desirable. While metal-catalyzed asymmetric reactions showed great ability in synthesis of chiral allenes,[6] organocatalytic asymmetric tools also demonstrated their practicability, but these organocatalytic methods still remain limited.[7,8] Most organocatalytic processes handled synthesis of di- or trisubstituted allenes,[7a–g] and those synthetic accesses to chiral tetrasubstituted allenes are rare.[7h–n,8] Among various approaches for chiral allenes, enantioselective functionalization of racemic allenes is an attractive and challenging one. Generally, α-hydrogen of allenoates and allene ketones has relatively low pK_a and displays certain acidity. In principle, it can be deprotonated and acts as a nucleophile. In 2013, Maruoka and co-workers reported alleno-Mannich and alkylation reactions of allenoates under phase-transfer catalysis.[8] In both reactions, the facile generation of cumulenolate under phase-transfer conditions was a key step. This pioneering work provided an concise method for synthesis of chiral tetrasubstituted allenes. With the use of allene anion as a nucleophile, we herein report an unprecedented organocatalytic C(sp^2)-H allylation of racemic trisubstituted allenoates with MBH carbonates to access axially chiral tetrasubstituted allenoates with good to excellent diastereo- and enantioselectivities (Scheme 1 c).

Previous work:

a) Tsuji-Trost allylation

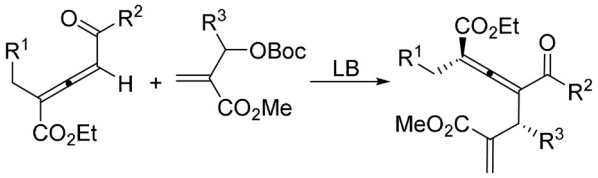


b) Lewis base (LB)-catalyzed allylation of MBH derivatives



This work:

c) asymmetric C(sp^2)-H allylation



Scheme 1. Strategies for asymmetric allylic alkylation.

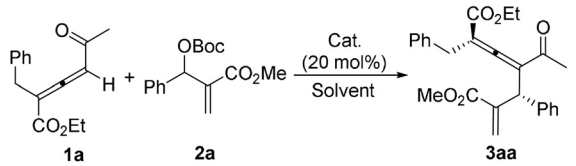
At the beginning of our study, the reaction of trisubstituted allenoate **1a** and MBH carbonate **2a** in CH_2Cl_2 was used as a model reaction to find the best catalyst (Table 1). The tertiary amine catalyst DABCO (1,4-diazabicyclo-[2.2.2]octane) showed excellent performance, and the product was obtained in 95 % yield in 0.3 h, although the diastereoselectivity was poor (entry 1). After getting these results, we next carefully evaluated the performance of various chiral Lewis bases, including cinchona-derived tertiary amines (**LB1**–**LB6**) and phosphines (**LB7**–**LB8**) with different scaffolds and functional groups. Quinine (**LB1**) only led to trace of product (entry 2). Other cinchona-derived tertiary amines **LB2**–**LB6** showed good catalytic activity, but both diastereoselectvity and enantioselectivity were not satisfactory (entries 3–7). Disappointingly, chiral phosphines (**LB7**–**LB8**) did not catalyze this reaction (entries 8–9). Compared with other catalysts, β-isocinchonine (**LB4**) was identified as the most effective catalyst in terms of catalytic activity and stereoselectivity, giving the product **3aa** in 89 % yield with

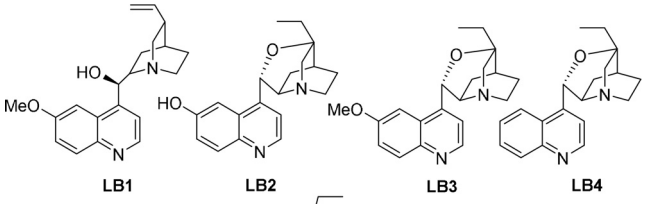
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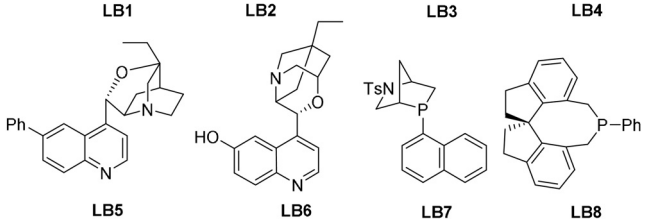
Supporting information and the ORCID identification number(s) for the author(s) of this article can be found under: https://doi.org/10.1002/anie.202009460.

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Table 1: Optimization of reaction conditions.[a]







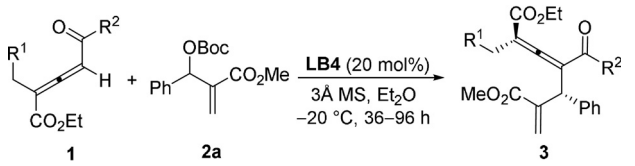
Entry	Cat.	Solvent	*t* [h]	Yield [%][b]	dr[c]	*ee* [%][d]
1	DABCO	CH_2Cl_2	0.3	95	1.6:1	–
2	**LB1**	CH_2Cl_2	48	trace	–	–
3	**LB2**	CH_2Cl_2	36	87	5:1	65
4	**LB3**	CH_2Cl_2	36	84	1:1	44
5	**LB4**	CH_2Cl_2	36	89	2:1	70
6	**LB5**	CH_2Cl_2	36	81	2:1	41
7	**LB6**	CH_2Cl_2	36	74	3:1	35
8	**LB7**	CH_2Cl_2	72	NR[g]	–	–
9	**LB8**	CH_2Cl_2	72	NR	–	–
10	**LB4**	toluene	48	85	10:1	10
11	**LB4**	THF	48	74	1:1	54
12	**LB4**	Et_2O	7	91	>20:1	84
13	**LB4**	MTBE	12	86	>20:1	83
14	**LB4**	DIPE	48	91	>20:1	82
15	**LB4**	1,4-dioxane	12	89	7:1	79
16[e]	**LB4**	Et_2O	48	80	>20:1	88
17[e,f]	**LB4**	Et_2O	48	82	>20:1	90

[a] Unless otherwise indicated, all reactions were performed with **1a** (0.1 mmol), **2a** (0.12 mmol) in 1 mL of solvent at 25 °C. [b] Isolated yield. [c] Determined by ^{1}H NMR. [d] Determined by chiral HPLC analysis. The *ee* value of the other diastereomer was between 5% and 13%. [e] The reaction was performed at −20 °C. [f] 3 Å MS (25 mg) was added as additive. [g] No reaction.

70% *ee*, albeit with poor diastereoselectivity (entry 5). Solvent screening (entries 10–15) revealed that diethyl ether was the optimal solvent, which resulted in 91% yield with 84% *ee* and >20:1 dr (entry 12). Using ethyl ether as the solvent, lowering the temperature helped slightly increasing *ee* value from 84% to 88% (entry 16). The addition of 3 Å MS also favored enantioselectivity (entry 17). All these results indicated that the optimal reaction conditions were: using 20 mol% of **LB4** in diethyl ether at −20 °C with 3 Å MS as an additive.

With the optimal reaction conditions in hand, we explored the substrate scope of trisubstituted allenoates **1** (Table 2). The asymmetric allylation reaction produced the tetrasubstituted allenoates in 78–99% yields with 17:1 to >20:1 dr and 90–92% *ee* (entries 1–14). The electronic and steric properties

Table 2: Substrate scope of trisubstituted allenoates.[a]



Entry	R^1	R^2	**3**	Yield [%][b]	dr[c]	*ee* [%][d]
1	Ph	Me	**3aa**	82	>20:1	90
2	2-FC_6H_4	Me	**3ba**	87	>20:1	90
3	3-FC_6H_4	Me	**3ca**	85	>20:1	90
4	4-FC_6H_4	Me	**3da**	88	>20:1	92
5	2-ClC_6H_4	Me	**3ea**	96	>20:1	91
6	4-ClC_6H_4	Me	**3fa**	81	>20:1	92
7	2-BrC_6H_4	Me	**3ga**	85	>20:1	92
8	4-BrC_6H_4	Me	**3ha**	83	>20:1	91
9	2-MeC_6H_4	Me	**3ia**	97	>20:1	90
10	4-MeC_6H_4	Me	**3ja**	78	>20:1	92
11	3-$MeOC_6H_4$	Me	**3ka**	99	>20:1	91
12	4-$NO_2C_6H_4$	Me	**3la**	89	>20:1	91
13	4-CNC_6H_4	Me	**3ma**	91	17:1	90
14	4-$CF_3C_6H_4$	Me	**3na**	90	>20:1	90
15	Ph	Et	**3oa**	90	>20:1	92
16	Ph	$BnCH_2$	**3pa**	73	>20:1	90
17	Ph	Ph	**3qa**	84	>20:1	91
18	*n*-Pr	Me	**3ra**	42	>20:1	90

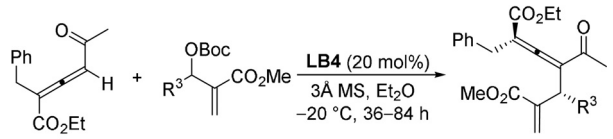
[a] Unless otherwise indicated, all reactions were performed with **1** (0.1 mmol), **2a** (0.12 mmol) and 3 Å MS (25 mg) in 1 mL of Et_2O for 36–96 h. [b] Isolated yield. [c] Determined by ^{1}H NMR. [d] Determined by chiral HPLC analysis.

of substituents on the phenyl groups of the allenoates had little effect on the reactivities and stereoselectivities (entries 1–14). In particular, the substrates with highly electron-deficient substituents on the benzene ring of the allenoates, including 4-$NO_2C_6H_4$, 4-CNC_6H_4 and 4-$CF_3C_6H_4$ substituted ones, afforded the corresponding products in 89–91% yields with 90–91% *ee* (entries 12–14). We next investigated the impact of the R^2 groups linked to ketone carbonyl on the reaction. Both alkyl groups such as ethyl, phenethyl and phenyl group in the allenoates had no remarkable impact on the reactivities and stereoselectivities, and all these allenoates performed well to give the tetrasubstituted allenoates in 75–90% yield with >20:1 dr and 90–92% *ee* (entries 15–17). When the R^1 group was an alkyl group, excellent diastereoselectivity and enantioselectivity were maintained, although the yield dropped (entry 18).

We next examined the reaction of trisubstituted allenoates **1a** with a variety of MBH carbonates **2** (Table 3). Diverse aryl-substituted MBH carbonates were compatible with the optimal conditions, and the reactions displayed high yields, excellent enantioselectivities and diastereoselectivities in most cases (entries 1–20). It appears that the position of the substituents on the phenyl ring of the R group did not significantly affect the reaction. The disubstituted substrates also worked to give the product **3ah** in 99% yield with >20:1 dr and 90% *ee* (entry 7). The substrates with 1-naphthyl and 2-naphthyl groups adapted well to the optimal reaction conditions and provided the corresponding products **3at** and

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Table 3: Substrate scope of MBH carbonates.[a]



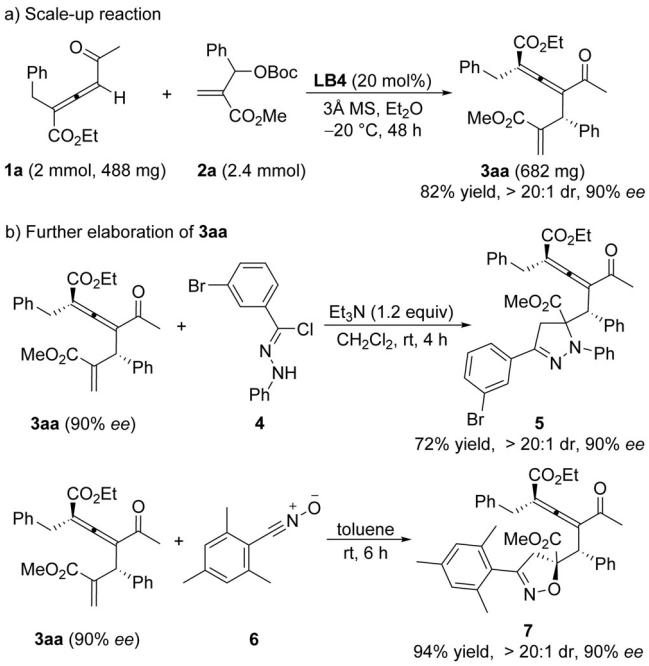
Entry	R^3	3	Yield [%][b]	dr[c]	*ee* [%][d]
1	2-FC_6H_4	**3ab**	91	>20:1	89
2	3-FC_6H_4	**3ac**	92	>20:1	94
3	4-FC_6H_4	**3ad**	90	>20:1	92
4	2-ClC_6H_4	**3ae**	85	17:1	89 (99)[e]
5	3-ClC_6H_4	**3af**	90	>20:1	93
6	4-ClC_6H_4	**3ag**	95	>20:1	92
7	3,5-$Cl_2C_6H_3$	**3ah**	99	>20:1	90
8	2-BrC_6H_4	**3ai**	95	>20:1	94
9	3-BrC_6H_4	**3aj**	92	>20:1	90
10	4-BrC_6H_4	**3ak**	88	>20:1	94
11	2-$NO_2C_6H_4$	**3al**	97	4:1	88
12	3-$NO_2C_6H_4$	**3am**	99	>20:1	90
13	4-$NO_2C_6H_4$	**3an**	98	>20:1	93
14	3-$MeOC_6H_4$	**3ao**	96	>20:1	90
15	4-$MeOC_6H_4$	**3ap**	90	>20:1	91
16	3-MeC_6H_4	**3aq**	91	>20:1	87
17	4-MeC_6H_4	**3ar**	89	>20:1	92
18	4-*i*-PrC_6H_4	**3as**	89	>20:1	90
19	1-naphthyl	**3at**	99	>20:1	90
20	2-naphthyl	**3au**	89	>20:1	94

[a] Unless otherwise indicated, all reactions were performed with **1a** (0.1 mmol), **2** (0.12 mmol) and 3 Å MS (25 mg) in 1 mL of Et_2O for 36–84 h. [b] Isolated yield. [c] Determined by ^1H NMR. [d] Determined by chiral HPLC analysis. [e] After recrystallization.

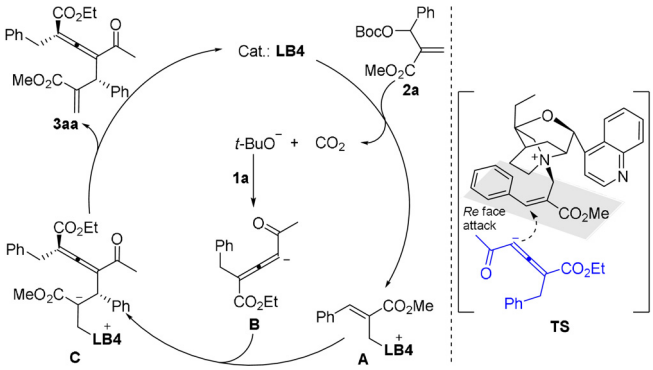
3au in 99% yield/90% *ee* and 89% yield/94% *ee*, respectively (entries 19–20). The absolute configuration of the products was unambiguously assigned through X-ray crystallographic analysis of the product **3ae**.[9]

As indicated in Scheme 2, the allylation reaction on the large scale proceeded smoothly to produce the product **3aa** in 82% yield with >20:1 dr and 90% *ee*. The tetrasubstituted allenoate products have several functional groups such as carbonyl, ester and carbon-carbon double bond, which provide a nice chance for further elaboration. Treatment of chiral allene **3aa** with nitrilimine **4** in the presence of Et_3N gave the pyrazoline derivative **5** in 72% yield with >20:1 dr and 90% *ee*. The cycloaddition of the chiral allene **3aa** with nitrile oxides **6** led to a dihydroisoxazole derivative **7** in 94% yield with >20:1 dr and 90% *ee*.[9]

As indicated in Scheme 3, a plausible mechanism for this asymmetric allylation was proposed. The addition of the catalyst **LB4** to MBH carbonate **2a** gave the intermediate **A** with simultaneous generation of *tert*-butoxy carbanion, which deprotonated trisubstituted allenoate **1a** to afford the allene carbanion **B**. Subsequent conjugate addition of the nucleophile **B** to the intermediate **A** led to formation of the tetrasubstituted allenoate intermediate **C**, which then produced the final product **3aa** through regeneration of the tertiary amine catalyst. In the transition state **TS**, due to bulky steric hindrance of the catalyst, the allene carbanion **B** attacks



Scheme 2. Scale-up reaction and further elaboration.



Scheme 3. A proposed reaction mehcanism.

the *Re* face of the olefinic bond in the intermediate **A**, which determines the absolute configuration of the product.

In summary, we have developed the first organocatalytic asymmetric C(sp^2)-H allylation of trisubstituted allenoates with MBH carbonates to access tetrasubstituted allenoates. Various trisubstituted allenoates and MBH carbonates were well tolerated under mild reaction conditions, providing novel tetrasubstituted allenoates with adjacent axial chirality and tertiary carbon stereocenters in high yields with good to excellent enantioselectivities and diastereoselectivities.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: alkylation · allene · allylation · asymmetric catalysis · organocatalysis

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